Artificial Intelligence and Decision Making (Course 6-4)

Department of Electrical Engineering and Computer Science

Bachelor of Science in Artificial Intelligence and Decision Making

General Institute Requirements (GIRs)

The General Institute Requirements include a Communication Requirement that is integrated into both the HASS Requirement and the requirements of each major; see details below.

Summary of Subject Requirements	Subjects
Science Requirement	6
Humanities, Arts, and Social Sciences (HASS) Requirement; at least two of these subjects must be designated as communication-intensive (CI-H) to fulfill the Communication Requirement.	8
Restricted Electives in Science and Technology (REST) Requirement [satisfied by 6.1200[J] and 18.C06[J] in the Departmental Program]	2
Laboratory Requirement (12 units) [satisfied by 6.1010 in the Departmental Program]	1
Total GIR Subjects Required for SB Degree	**17**

Physical Education Requirement
Swimming requirement, plus four physical education courses for eight points.

Departmental Program

Choose at least two subjects in the major that are designated as communication-intensive (CI-M) to fulfill the Communication Requirement.

Fundamentals		Units
Choose 12 units of the following:		12
6.1000	Introduction to Programming and Computer Science	
6.100A & 6.100B	Introduction to Computer Science Programming in Python and Introduction to Computational Thinking and Data Science	
6.100A & 16.C20[J]	Introduction to Computer Science Programming in Python and Introduction to Computational Science and Engineering	
6.1010	Fundamentals of Programming	12
6.1200[J]	Mathematics for Computer Science	12
6.1210	Introduction to Algorithms	12
18.C06[J] or 18.06	Linear Algebra and Optimization Linear Algebra	12
Select one of the following:		12
6.3700	Introduction to Probability	
6.3800	Introduction to Inference	
18.05	Introduction to Probability and Statistics	
Centers		
Select five subjects, including one from each area:		60
Data-centric		
6.3720	Introduction to Statistical Data Analysis	
6.3900	Introduction to Machine Learning	

6.C01 & C011	Modeling with Machine Learning: from Algorithms to Applications and Modeling with Machine Learning for Computer Science	
Model-centric		
6.3000	Signal Processing	
6.3100	Dynamical System Modeling and Control Design	
6.4110	Representation, Inference, and Reasoning in AI [1]	
6.4420[J]	Computational Design and Fabrication	
6.4400	Computer Graphics [2]	
Decision-centric		
6.3100	Dynamical System Modeling and Control Design	
6.4110	Representation, Inference, and Reasoning in AI [1]	
6.C571[J]	Optimization Methods [3]	
Computation-centric		
6.1220[J]	Design and Analysis of Algorithms	
6.1400[J]	Computability and Complexity Theory	
6.4400	Computer Graphics [2]	
6.C571[J]	Optimization Methods [3]	
Human-centric		
6.3260[J]	Networks	
6.3950	AI, Decision Making, and Society	
6.4120[J]	Computational Cognitive Science	
6.4590[J]	Foundations of Information Policy	
6.C35[J]	Interactive Data Visualization and Society	
Communication-intensive in the Major		
Select one of the following Application CI-M subjects:		12-15
6.4200[J]	Robotics: Science and Systems (CI-M)	
6.4210	Robotic Manipulation (CI-M)	
6.8611	Quantitative Methods for Natural Language Processing (CI-M)	
Electives		
One subject that satisfies a degree requirement in Course 6 or Course 18		12-15
One subject from the list of AI+D Advanced Undergraduate Subjects		12-15
Students must satisfy at least one program requirement or elective with a second CI-M subject (beyond the application CI-M requirement) from the department's CIM2 list.		
Social and Ethical Responsibilities of Computing (SERC)		
Students must satisfy at least one program requirement or elective with a subject from the SERC-qualified subjects listed below.		
Units in Major		**168-177**
Unrestricted Electives		**48**
Units in Major That Also Satisfy the GIRs		(36)
Total Units Beyond the GIRs Required for SB Degree		**180-189**

The units for any subject that counts as one of the 17 GIR subjects cannot also be counted as units required beyond the GIRs.

[1] *6.4110 Representation, Inference, and Reasoning in AI may count toward models or decision-centric, but not both.*

[2] *6.4400 Computer Graphics may count toward model-centric or computation-centric, but not both.*

[3] *6.C571[J] Optimization Methods may count toward decision-centric or computation-centric, but not both.*

[4] *Subject has prerequisites that are outside the program.*

AI+D Advanced Undergraduate Subjects

6.3020[J]	Fundamentals of Music Processing	12
6.3730[J]	Statistics, Computation and Applications	12
6.4210	Robotic Manipulation	15
6.4300	Introduction to Computer Vision	12
6.5151	Large-scale Symbolic Systems	12
6.5831	Database Systems	12
6.5931	Hardware Architecture for Deep Learning	12
6.7411	Principles of Digital Communication	12
6.8371	Digital and Computational Photography	12
6.8611	Quantitative Methods for Natural Language Processing	15
6.8701[J]	Computational Biology: Genomes, Networks, Evolution	12
6.8711[J]	Computational Systems Biology: Deep Learning in the Life Sciences	12
6.8801[J]	Biomedical Signal and Image Processing	12
18.404	Theory of Computation	12

Social and Ethical Responsibilities of Computing (SERC) Subjects

6.3900	Introduction to Machine Learning	12
6.3950	AI, Decision Making, and Society	12
6.4300	Introduction to Computer Vision	12
6.4590[J]	Foundations of Information Policy	12
6.8611	Quantitative Methods for Natural Language Processing	15
6.C01	Modeling with Machine Learning: from Algorithms to Applications [1]	6
6.C40[J]	Ethics of Computing	12

[1] *Credit cannot be awarded without simultaneous completion of a 6-unit disciplinary module. Consult advisor.*